

SCALE-INVARIANT SALIENT EDGE DETECTION

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ABSTRACT

In this paper, we propose a Scale-Invariant Salient Edge Detection framework (SISED) using Hadamard-Product operation. Current HED-like edge detection approaches fuse multiple side outputs to produce the final edge map, which contains noise and unwanted edge details. Scale-invariant salience provides strong and convincing evidence for precisely describing object contour. The normalized Hadamard-Product is able to find and extract scale-invariant features by multiplying a set of side outputs. The computed Scale-Invariant Salient Edge (SISE) maps capture the hierarchical structure of contour details and can be utilized to improve the detection accuracy. The experimental results show SISED reaches state-of-the-art performance.

Index Terms— Scale-invariant, Edge detection, Hadamard-Product, Deep network

1. INTRODUCTION

Human visual perception largely relies on shapes [1]. Contour-based shape features provide compact yet expressive descriptions about object shape salience and play an important role in visual analysis tasks such as image segmentation, object detection, and recognition. Traditional edge detection methods exploit low-level visual cues to construct hand-crafted features then classify the edge and non-edge pixels using threshold-based methods [2]. The results of conventional approaches lack semantics at the object level. Recently, convolutional neural network (CNN)-based approaches have become mainstream in the image processing domain. Among deep network-based edge detection methods, HED [3] is one of the successful frameworks. HED produces five intermediate side outputs (SO) (SO-1 to SO-5 in Fig.1(c-g)) and performs deep supervision along the VGG16 network pathway. Its final fused result achieves performance within a 2% gap to human vision. Since then, several approaches use a similar architecture to further improve the accuracy. These efforts mainly focus on improving the quality of side outputs or enhancing the deep supervision strategies. For example, RCF[4] uses extra convolutional layers to embed finer details into the side outputs. CED[5] adds a top-down backward-refining pathway to increase the crispness of side

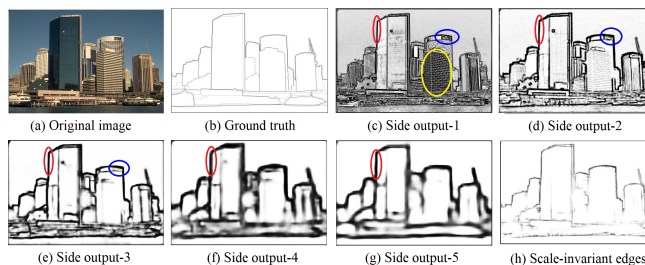


Fig. 1. Five side outputs (c-g) from the enriched backbone network and the extracted scale-invariant salient edges (h). One can clearly see that scale-invariant edges present very similar contents to the ground truth (b).

outputs using subpixel convolution. RDS[6] applies relaxed labels for supervising intermediate layers. DeepBoundary[7] applies multi-resolution learning with more training samples. These approaches push the edge detection accuracy to outperform the human vision performance. More recently, different network structures like Generative Adversarial Network (GAN) is used in the edge detection field to further improve the performance[8].

However, semantic edge detection still remains challenging. Each SO contains edge maps with a certain spatial scale. Finer scale SOs have both thin contours and noise while coarser SOs have fatty edges from upsampling lack necessary details. For example, in the fine scale SO-1 (Fig.1(c)), boundaries circled in red appear in all side outputs, some (circled in blue) are just in SO-2 and SO-3 only, and clutters (circled in yellow) only exist in SO-1. Intuitively, thin edges with strong scale expression (scale-invariant across more scale levels) have higher capacity of describing shapes and should be part of the final edges. The problem with the existing fusion process is that each side output's weight is applied indiscriminately without considering hierarchical edge importance. This poses the dilemma to the network: to include desired features it has to accept many unwanted data and vice versa. Consequently, the result often contains more noise and thick edges while missing some key boundaries. To tackle this issue, a complicated deeper network is required to learn accurate pixel-wise importance, which is often difficult due to gradient vanishing/exploding and a shortage of labeled data.

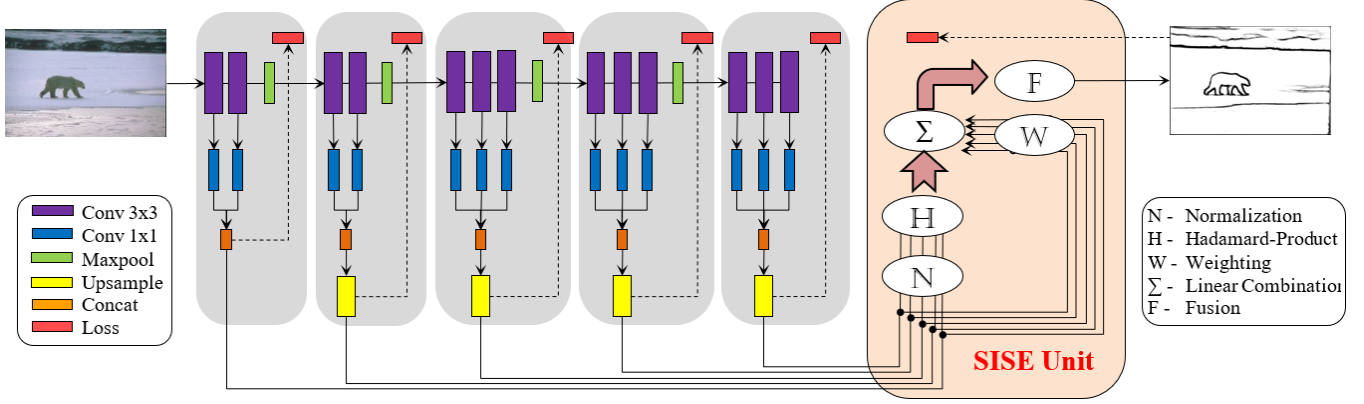


Fig. 2. SISED Network Architecture.

Instead, we introduce a Scale-Invariant Salient Edge Detection (SISED) network to locate and extract the important Scale-Invariant Salient Edge (SISE) as the subset of each side output without increasing the network complexity. The normalized Hadamard-Product is the key operation of SISED where a multiplicative operation is applied to promote mutually agreed features across multi-scale side outputs while suppressing those with weak scale expression. Fig. 1(h) presents an example SISE map of Scale-invariant thin edges across all side outputs and closely resembles the ground truth (Fig. 1(b)). SISED computes a set of SISE maps representing the edge importance hierarchically which are further utilized to enhance the edge results. The experiments conducted on the BSDS500 [9] and NYUD[10] datasets show that SISED is effective and reaches state-of-the-art performance.

2. OUR APPROACH

The architecture of the proposed framework is shown in Fig. 2, where the Hadamard-Product is the key operation inside the novel SISE Unit.

Normalized Hadamard-Product

The Hadamard-product itself is an element-wise multiplication of two matrices. If values of both matrices are normalized $\in [0, 1]$, it naturally results in an AND gating operation. Fig. 3 presents the surface plot of a multiplicative function

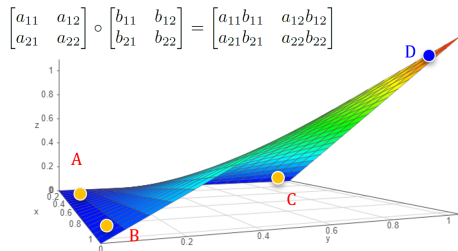


Fig. 3. Normalized Hadamard-Product

$z = x \cdot y, x, y \in [0, 1]$. Point A's z is minimum because both A's x and y are small. The z values of points B and C are minimal because either x or y of these two points are small. Only Point D's z value is maximized due to both x and y being large. This AND gating operation of the normalized Hadamard-Product acknowledges the salience if it is mutually supported by both inputs, and its result is scale-invariant if the inputs represent different scales. The normalized Hadamard-Product (Eq. 2) is a differentiable function with a smooth surface, which is a desired property for gradient-based convex optimization in deep learning networks.

Scale-Invariant Salient Edge Detection (SISED)

As seen in Fig. 1, some edge features (e.g. in c-d) contain cluttered edges which are not important to describe object shapes while boundaries in (c)-(g) are fatty due to upsampling. Fig. 1(h) presents object contours by providing thinner edges agreed by multiple scale outputs. These scale-invariant thin edges are resilient to transformation changes and salient to human visual perception. They can capture the image gist and contribute more to visual recognition tasks. Our goal is to find and extract those SISE to improve the edge detection quality.

As shown in Fig. 2, SISED consists of the VGG16-based backbone network with five stages and the novel SISE unit. Similar to RCF, each stage has extra convolutional layers to enrich the edge features. The SISE unit mainly has the following operations before fusion: normalization, Hadamard-Product, and linear combination.

Normalization: It is important for the Hadamard operation to act as an AND gate to boost mutually agreed scale-invariant features and suppress disagreed edges. The data of side outputs is in a wide range, e.g. data from SO-1 is $\in (-200, 500)$. To avoid incorrect AND gate behavior from multiplying two negatives, a sigmoid function is used to normalize the data $\in [0, 1]$:

$$f_i = \frac{1}{1 + \exp(-so_i)} \quad (1)$$

where so_i is a side output, $i \in [1, 5]$, and its values could be in $(-\infty, +\infty)$; the normalized f_i is between 0 and 1.

Hadamard-Product: By applying Hadamard (element-wise) product, a SISE map is obtained across multiple normalized side outputs:

$$SISE_1^k = \prod_{i=1}^k f_i \quad (2)$$

where f_i is the normalized SO_i from Eq. 1 and k indicates the number of side outputs involving the calculation. Fig. 4 shows examples of computed SISE maps. The left column contains five side outputs. The middle column has $SISE_1^2 \sim SISE_1^5$ from top to bottom, where scale-invariant edges are presented hierarchically. Compared with the left column, edges in SISE maps are much thinner and reduced significantly because only scale-invariant features are kept due to their high mutually agreed values while disagreement between these maps is diminished. As k increases, edges in SISE maps decrease as well because less edge detail survives when more scale views are examined. $SISE_1^5$ contains a clearer view than $SISE_1^2$ because it considers the scale view from each side output. $SISE_1^5$ has almost all ground truth contours without many cluttered pixels.

Linear combination: Hierarchical SISE maps capture overall object boundary structures and provide edge importance at multiple levels. Their probability-like values $\in [0, 1]$ can be used in linear activation-like functions to boost strong edges and suppress non-essential pixels. Eq.3 integrates the side outputs with SISE to enhance the final results:

$$F_i = \alpha SO_i + W \cdot SISE \quad (3)$$

where SO_i is the original side output and W are a set of learned weights for hierarchical $SISE_1^k$ maps. Intuitively after the combination, feature values with high SISE values are amplified, while others are still preserved or diminished proportionally according to the learned weights and SISE. This approach not only extracts and promotes essential scale-invariant edges but also preserves detail.

Network Training and Loss Functions

Each stage in SISED network is trained with layer-specific side supervisions. The final fused result is supervised at the last loss layer as well. The overall loss is formulated as,

$$L = \sum_{k=1}^5 w_k \cdot L_k(P, Y) + w_{fuse} \cdot L_{fuse}(P, Y) \quad (4)$$

where w_k and w_{fuse} are weights for the side loss and fusion loss respectively. L_k and L_{fuse} functions compute the difference between edge prediction P and the edge label Y :

$$L(p_i, y) = -\alpha \sum_{i \in Y_-} \log(1 - p_i) - \beta \sum_{i \in Y_+} \log(p_i) \quad (5)$$

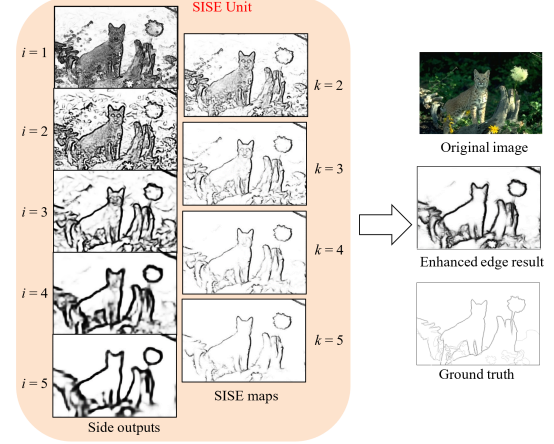


Fig. 4. Hierarchical SISE maps for edge enhancement. The left column contains side outputs ($i = 1 \sim 5$) while $SISE_1^k$ maps ($k = 2 \sim 5$) are in the middle.

where p_i is the edge prediction and Y_+ and Y_- denote edge and non-edge pixels respectively. Both $\alpha = \lambda \cdot |Y_+| / (|Y_+| + |Y_-|)$ and $\beta = |Y_-| / (|Y_+| + |Y_-|)$ are used to balance the training samples and are controlled by the hyper-parameter λ . It is also worth noting that all the steps inside the SISE unit (sigmoid, Hadamard-Product, and linear combination) are differentiable which means derivatives exist at all points. Thus, during network backpropagation no additional adjustment is required for parameter learning.

3. EXPERIMENTS

Datasets and Implementation

We evaluated the proposed approach on two public datasets: BSDS500 [9] and NYUD[10]. BSDS500 contains 200 training images, 100 validation images, and 200 testing images. The PASCAL VOC Context dataset [11] was added into BSDS500 as the additional training set. The ground truth is the averaged annotation labeled manually by multiple annotators. NYUD has 1449 pairs of aligned RGB and depth images which are split into 381 training, 414 validation, and 654 testing. To increase the training data size, all training or validation images in both datasets were augmented by randomly flipping, scaling, and rotating operations. The network is implemented using PyTorch. The pre-trained VGG16 model is used as the network initialization. Regarding the loss functions, w_k and w_{fuse} are set as 0.5 and 1.1 respectively, and the hyper-parameter λ is set as 1.1 and 1.2 for BSDS500 and NYUD respectively. SGD optimizer is used to train the network for 40k iterations on both datasets. The batch size is set to 10 for all the experiments. The initial learning rate, momentum, and weight decay are set to 1e-6, 0.9, and 2e-4 respectively. The learning rate decreases by 10 times after every 10k iterations.

Ablation Study

We examine the quality of individual side outputs and $SISE_1^k$ maps by conducting training and testing on BSDS500. Tab.1(a) shows the detection performance of individual SO_i without fusion. Note, $i = 1$ means SO_1 is the final edge map. In general, the qualities of SOs from later stages are better due to the diminishing details and noise. SO_4 is the most accurate as it finds the balance between global contours and fine details. Tab.1(b) shows the comparison among hierarchical $SISE_1^k$ maps. Larger k means scale-invariant edges are endorsed by more scale views. $SISE_1^1$ is identical to SO_1 , while $SISE_1^5$ contains the best edge quality by including scale-invariant salient edges across all five stages. Note, without fusion, $SISE_1^5$'s ODS has reached .775, only .027 less than human vision. This indicates the SISE maps perform well in capturing important edge features.

Table 1. Performance of individual SO_i and $SISE_1^k$ on BSDS500.

i	ODS	OIS	AP	k	ODS	OIS	AP
1	.608	.634	.617	1	.608	.634	.617
2	.713	.733	.709	2	.692	.714	.711
3	.761	.781	.778	3	.736	.757	.742
4	.770	.784	.763	4	.762	.783	.752
5	.747	.754	.702	5	.775	.789	.717

(a) SO_i

(b) $SISE_1^k$

Comparison with State-of-the-Arts.

We conducted comparative experiments on BSDS500 dataset. Fig.5(a) show the evaluation results. The performance of the human eye in edge detection is known as 0.803 ODS F-measure. Our method achieved better than humans, and its precision-recall curve is higher than many including HED and RCF-MS[3, 4]. These results demonstrate our scale-invariant edge features are effective and robust. From Tab. 2, SISED ranks at the top for ODS F-measure. Its ODS is 2.7%, 0.4%, and 2.3% higher than HED, RCF-MS, and RDS respectively. In OIS and AP, we achieved similar performance gain. We also conducted comparisons on NYUD dataset. Fig. 5(b) shows the precision-recall curve of SISED is higher than others. In Tab.3, SISED ranks higher than all others with an OIS F-measure 2.9% higher than HED and 1.5% higher than RCF.

4. CONCLUSION

Existing fusion processes do not differentiate the side output's internal pixel importance and as a result causes noisy edges and missing boundaries. We propose a Scale-Invariant Salient Edge Detection (SISED) network to tackle this issue. The key idea is to utilize the normalized Hadamard-Product to perform an AND gating operation across multi-scale side outputs

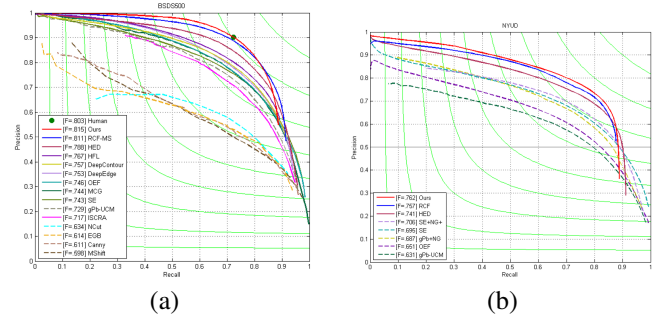


Fig. 5. The precision-recall curves of our method and other works on BSDS500 (a) and NYUD test set (b).

Table 2. The comparison with some methods on BSDS00.

Method	ODS	OIS	AP
Canny[2]	.611	.676	.520
EGB[12]	.614	.658	.564
MShift[13]	.598	.645	.497
OEF[14]	.746	.770	.815
HFL[15]	.767	.788	.795
N^4 -Fields[16]	.753	.769	—
DeepContour[17]	.757	.776	.790
DeepEdge[18]	.753	.772	.807
RDS[6]	.792	.810	—
CEDN[19]	.788	.804	—
HED[3]	.788	.808	.840
RCF-MS[4]	.811	.830	.846
Our SISED	.815	.835	.839

Table 3. The comparison with some methods on NYUD.

Method	ODS	OIS	AP
gPb-UCM[9]	.631	.661	.562
OEF[14]	.651	.667	.653
gPb-NG[20]	.687	.716	.629
SE[21]	.695	.708	.719
SE+NG[22]	.706	.734	.549
HED[3]	.741	.757	.749
RCF[4]	.757	.771	.749
Our SISED	.762	.786	.752

along the network pathway. The computed SISE maps contain scale-invariant features that provide the hierarchical structure about edge detail. Integrated with the original side outputs, SISE maps guide the network to learn the semantic edge boundaries. Ablation studies and comparative experiments show that the proposed approach is effective and achieves state-of-the-art performance.

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